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Automating actions in a distributed computer system

Abstract

The present disclosure provides systems and methods performed in a central system. An initial entity model comprising a plurality of artefacts is formed. A runtime engine determines a plurality of available agent engagements based on the initial entity model. Responsive to an input selecting one of the available agent engagements, a notification is sent to an external autonomous agent. An indication of an action to be performed is received from the external autonomous agent. Responsive thereto, a message is sent to an external service indicating the action, thereby causing the action to be performed. The entity model is updated with a new artefact corresponding to the performed action, and a further plurality of agent actions is determined based on the updated entity model.

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Background/Summary

FIELD

(1) The present disclosure pertains to mechanisms for automating system actions in a distributed computer system.

BACKGROUND

(2) Automation in the field of artificial intelligence (‘AI’) has increasingly emphasized large, deep

machine learning (ML) models. Much recent focus has been given to ‘general’ intelligent models that can perform a wide range of tasks on which they have not been explicitly trained. Such models may exhibit emerging behaviors that are hard to predict.

SUMMARY

(3) By contrast, an extensive and modular intelligence architecture is considered herein, providing an advanced form of expert system. Specific expertise and capabilities are provided by autonomous agents in a manner that is much more tightly controlled and predictable than current large models such as large language models (LLMs). A central system coordinates sophisticated collaborative decision automation for an entity such as an organization using sophisticated and explainable entity modelling. The system is highly scalable and can accommodate entities of arbitrary complexity.

(4) In various aspects herein, a distributed architecture comprises a central system, one or more external autonomous agents, and one or more external services. The central system interfaces with the external autonomous agents and the external services, e.g. via suitable interfaces such as programmatic interfaces or APIs. The central system stores a shared entity model which can serve as a communication model between the multiple external agents, enabling the agents to collaboratively solve problems, formulate strategies/plans, and then issue actions via the central system to external services. Inputs from external agents may be passed to the external services. Actions may originate externally to the central system. An agent may identify an action and cause a modification to the model. Based on the updated context (in view of the modification to the model), the runtime engine may inspect the context, and notify agents of the updated context such that the agent may undertake actions as part of their role.

(5) The central system provides an interface or ‘wrapper’ between the agents and the services. In one example, the central systems hold a credential or set of credentials that enables it to authenticate with each external service. The agents are authenticated with the central system, and can only perform actions on the external services via the central system.

(6) The central system contains an entity model, which is used as a basis for engaging the external agents. As actions are performed, the entity model is updated, which in turn may reveal new options for agent engagement. Thus, as the entity model is updated as actions are performed, new options for agent engagements become available, leading to further actions.

(7) Explicit entity models are considered, which are built from defined ‘artefacts’. In one example, artefacts are implemented as instances of artefact classes. In such cases, an artefact is analogous to an object in object oriented programming (OOP). An artefact class programmatically describes some aspect of an entity and, where applicable, its possible relationship(s) to other artefact(s).

(8) One example application is process automation in an organization based on an explicit model of the organization.

(9) A first aspect herein is directed to a computer-implemented method performed in a central system, the method comprising: forming in electronic storage, an initial entity model comprising a plurality of artefacts; determining, by a runtime engine, a plurality of available agent engagements (options) based on the initial entity model; responsive to an input selecting one of the available agent engagements, sending a notification to an external autonomous agent; receiving from the external autonomous agent an indication of an action to be performed, and responsive thereto: sending a message to an external service for causing the action to be performed, updating the entity model with a new artefact corresponding to the performed action; and determining a further plurality of agent actions based on the updated entity model.

(10) The method may be repeated in an iterative fashion as actions are performed, and new agent engagements become available as the model is updated.

(11) In embodiments, each artefact may be generated as an instance of an artefact class defined in an artefact library, the plurality of available agent engagements and the further plurality of agent actions extracted from the artefact library using the initial entity model and the updated entity model respectively.

(12) In some examples, the input selecting one of the available agent engagements may be received from a user interface based on a selection made by a human user.

(13) In some examples, the instruction selecting one of the available agent engagements may be received from an external agent engaged to perform a role defined in an artefact of the initial model. In this manner, a first external agent that is notified of a change in the entity model can request (or suggest) engagement of a second agent.

(14) In some examples, determining a further plurality of agent actions based on the updated entity model comprises: inspecting the updated entity model, determining that the updated entity model is in a state that satisfies a criterion defined in a library, and notifying one or more autonomous agent of the state of the entity model.

(15) The central service may provide a registration function enabling external agents to be registered with the central system. An external agent is registered by associating a defined role with an agent identifier of the external agent. Registering a role in this manner indicates the external agent is able to fulfil that role. The plurality of available agent engagements may be determined based on a set of roles that have been associated with external agents. When an additional agent is registered with the system, the available agent engagements may be updated in response, even if the entity model has not changed at that point.

(16) The artefact library defines possible artefact types and possible agent engagements. The library contains applicability rules, which determine when each agent engagement is available. An agent engagement becomes available when a particular type of artefact or combination of artefacts is present in the entity model. The entity model may additionally include connections (links) between artefacts representing relationships between them, and the applicability rules may depend on the artefacts and links that are present (e.g., a certain agent engagement may become available when a specific set of artefacts are connected in specific way). A set of artefacts and links therebetween, and agents engaged to perform roles defined by artefacts, and the states defined in a library may together be referred to as a 'context'. The library may define contexts (states of artefacts and links/relationships) wherein a role becomes relevant, and an agent may be notified to undertake their role when such a context is identified in the model (e.g., through model inspection by the runtime engine).

(17) This architecture means the library (or libraries) in the central system can be scaled in anticipation of future agent capabilities. New automation functionalities emerge dynamically as new contexts become available and/or new external agents become available, opening up new agent engagement possibilities, causing the model to grow as actions are performed.

(18) A second aspect herein is directed to a computer-implemented method performed in a central system, the method comprising: forming in electronic storage an entity model comprising: a plurality of artefacts, each artefact being an instance of an artefact class defined in an artefact library, and one or more links between the artefacts; inspecting, by the runtime engine, the artefact library based on the entity model to determine a set of available agent engagements, the artefact library defining possible agent engagements and applicability rules pertaining thereto; responsive to an input selecting one of the available agent engagements, sending a notification to an external autonomous agent; receiving from the external autonomous agent an indication of an action to be performed, and responsive thereto, sending a message to an external service for causing the action to be performed.

(19) In embodiments, the artefact library may specify an agent capability for each possible agent engagement, and the central system may select the external agent by matching the external agent with the specified agent capability. This provides a modular architecture, whereby new agents with relevant capabilities may be introduced without modification to the underlying artefact library.

(20) Embodiments of the first or second aspect may comprise the central service sending with the message with a credential (such as an authentication token or other proof of identity) for authorizing the action, the credential held in the central system in a storage location inaccessible to

the external autonomous agent.

(21) In some examples, the instruction selecting one of the available agent engagements may be received from a user interface based on a selection made by a human user. In some examples, the instruction selecting one of the available agent engagements may be received from an external agent engaged to perform a role defined in an artefact of the initial model.

(22) The library may associate roles with contexts before agents that can fulfil those roles become available. New agent engagements may subsequently become available when an external agent registers itself for a role defined in the library that was not previously available.

(23) Further aspects relate to computer system programmed to implement any of the above methods, and computer-readable instructions for programming a computer system to implement the same.

(24) The initial entity model may be formed in computer memory, or in separate electronic storage.

(25) One of the challenges of automating entities is that entities in a given problem domain (e.g., businesses in a particular sector) appear to have similar structures but in reality the model of any particular business will be extremely specific depending upon the lifecycle phase they are in and the myriad choices of suppliers, vendors, services, practices, etc. This leads to a need to have highly customizable entity models which share as much commonality as possible such that expert agents can apply their analytical skills to specific sub-models of the business and work collaboratively with other agents towards high-level objectives. Furthermore, this demands the ability to specialize models on a per-case basis and extend the available features.

(26) Examples herein further enable extensions and improvements contributed by third party experts to be incorporated into the platform, thus providing an adaptable technical framework for automating entities of various forms, sizes, and in various problem domains.

(27) Examples described herein enable context-informed decision making and enable real-world actions to be performed by and on behalf of an entity. Any number of different actions may be automated, so long as an external agent configured to perform the action is available to be engaged by the central system. Advantageously, the examples herein enable automation of processes in a technical system, and the processes in the technical system are configured to cause real-world outputs and actions to be carried out.

(28) Example embodiments will now be described in further detail.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a better understanding of the present disclosure, and to show how embodiments of the same may be carried into effect, reference is made by way of example only to the following figures in which:

(2) FIG. 1 shows an example system;

(3) FIG. 2 shows an example decision infrastructure;

(4) FIG. 3 shows a block diagram representing model-building techniques;

(5) FIG. 4 shows a block diagram of an example model;

(6) FIG. 5A shows an example AI system;

(7) FIG. 5B shows an example AI system including a cognitive layer;

(8) FIG. 6 shows a flowchart of an example sequence of agent actions, taken based on a model context;

(9) FIG. 7a shows an example structure of an initial entity model;

(10) FIG. 7b shows an example user interface for selecting an artefact option to add to a model;

(11) FIG. 7c shows an example user interface for selecting an agent engagement option;

(12) FIG. 7d shows the model of FIG. 7a, updated to include additional artefacts;

- (13) FIG. 8 shows an example system in communication with an external agent;
- (14) FIG. 9 shows a flowchart of an example dynamic context matching process;
- (15) FIG. 10 shows an example entity model updating process resulting from a context matching process;
- (16) FIG. 11 shows an example model structure and associated libraries; and
- (17) FIG. 12 illustrates an example process of instantiating artefacts in an entity model.

DETAILED DESCRIPTION

- (18) Tools and systems described herein enable process automation based on an explicit model of an entity. In the described embodiments, a central system may be referred to as a 'Honu' platform or simply as Honu.
- (19) Developments in artificial intelligence (AI), including recent developments in generative AI, have enabled automation across many sectors. However, the inventors have noted that automating higher level structures such as entities like entire organizations may not be effective by merely automating aspects and processes of the organization separately in a disjointed fashion. The description herein provides examples of tools and techniques for modeling parts of an organization, including links therebetween, in order to automate processes in view of a wider context of all components in the entity.
- (20) With reference to FIGS. 5a and 5b, the inventors have noted that systems wherein large AI tools 510 are in direct communication with data 520 to make decision (as in FIG. 5a) may not be as effective as systems using 'smaller' AI components (e.g., 510a, 510b, 510c) implementing separate functionalities. The AI components 510a-c may access and leverage data based on a model and in view of a context provided by a cognitive layer 530. Note that a 'small' AI component could be a large model (such as an LLM) but assigned a targeted role, task etc.
- (21) The cognitive layer 530 may act like a nervous system for the entity being modeled, equipping AI agents with the ability to understand context of the roles/actions they are performing.
- (22) As discussed later herein, agents may be engaged to perform a role, where that role is defined as an artefact in a model of the entity. The techniques described herein provide a technical framework for modelling an entity, and for enabling agents to perform their roles based on the model.
- (23) In some examples, agents may automatically take on certain tasks as part of the roles they have been engaged to perform. The techniques described herein provide a technical framework for modelling an entity, including features thereof and links therebetween, and for enabling agents to perform their roles based on the model.
- (24) A model of an entity such as a business may be built from libraries from a specific domain. Agents may dock onto the model, interacting with the business and other agents as well as one or more external service. Libraries may refer to specific packages of stylized program code such as python code. Libraries define artefact classes which are used in creating entity models.
- (25) The models described herein are dynamic and may update based on changes to the entity. Further, the model may update based on contexts identified within the model (e.g., particular structures of artefacts), and changes to the entity in the real world may be effected based on the change to the model, e.g., via an external agent instructed by the central system to act on behalf of the entity. Dynamic context matching processes are described in more detail later herein.
- (26) The diagram of FIG. 6 represents an example emergent automated process, which may be automated based on the modelling and agent engagement techniques discussed herein. The emergent process may not be explicitly defined in the libraries or code. However, the dynamic matching of contexts and expertise produces a behavior, which post-hoc in isolation, may be described as a process. Moreover, techniques described herein generate an automated process by building and layering contexts for an entity, and enabling agents to communicate and act using the model. The automated process is not defined explicitly, but arises based on the context. FIG. 6 is described in more detail later. Examples of systems and techniques for implementing dynamic

context matching are also provided later herein.

(27) FIG. 1 shows components of a system for configuring a model of an entity and for engaging agents to take actions based on the model.

(28) A central console **110** provides an interface to a platform for building models of entities. In some examples, organizations and businesses associated with the entities may be modeled. The console **110** may be an application developed to configure models of entities such as businesses, and to see metrics and run simulations based on the model. The model acts as a digital twin of the entity being modelled. The components denoted by reference numerals **110**, **120**, and **122** together comprise the platform.

(29) In the example of FIG. 1, the central console enables a business owner to access user interface (UI) application components **112** providing functionalities for building a model, activating integrations with external services, performing analytics and simulations based on the model, accessing a store of registered agents, and accessing libraries of components for inclusion in the model, and monitoring installs, usage, billing and other metrics of a given agent. The explicit components of a model may be referred to as artefacts.

(30) A second example set of UI application components is shown in FIG. 1, representing similar tools but external agent providers to develop tools for implementing roles that may be modelled. The developer console may be an application developed to allow developer partners to manage and monitor agents that they configure. Developer partners may be freelancers or software providers. The developer partners may make their services available or monetize expertise in the form of an agent on the platform.

(31) The interface for building models may comprise no-code interfaces, such as graphical tooling, guided interfaces etc.

(32) An agent UI may be configured to provide an alternative interface to the agent itself.

(33) Business owner accessing the business console and associated UI tools **112** may configure the model through a developer, or may act themselves to build a model. A business owner in context of FIG. 1 may be someone who owns or is acting on behalf of a business.

(34) The console **110** operates through an application programming interface (API) **120** to communicate with internal platform services **122** including, for example, model runtime, library registry, service registry, agent registry, business mechanics tools, and a model template registry.

(35) Model runtime may refer to a sandboxed environment wherein a model is run. The model may be run locally on the platform infrastructure, where the platform makes calls to engaged agents and receives action invocations proposing actions to take (i.e., roles to fulfil).

(36) The library registry refers to a catalog of available libraries which may be imported for use in constructing a model. The library registry may provide an API to register and unregister libraries.

(37) The agent registry refers to a catalogue of available agents which may be engaged to fulfil roles that are encoded in a model. The agent registry may provide an API to register and unregister agents. Agents may register roles via the system. After an agent registers a role, an option to engage that particular agent to perform the role for which it is registered may be provided when building an entity model, provided the model has a context in which that role is relevant.

(38) The service registry refers to a catalogue of available services which may be encoded as artefacts in a model and leveraged by agents that are engaged to fulfil roles via the service. The service registry may provide an API to register and unregister services.

(39) Business mechanics refers to a representation of the relationships amongst different model artefacts. These objects may be used to run simulations and for forecasting.

(40) The model template registry may be a catalogue of available template models, which may be added to a model to quickly build larger and more complex models. The model template registry may provide an API to register and unregister templates.

(41) The following description provides definitions of terms that may be referred to herein.

(42) A domain may refer to a collection of models that share a common set of libraries.

- (43) A model will be understood as a representation of an entity such as a business or organization. The model represents relationships between components and elements of the entity. A model may exist within a domain using artefacts declared in the libraries which may be instantiated at runtime.
- (44) An artefact is an instance of a class which is implemented in the model to represent an aspect of the entity. A particular artefact may be a unique combination of the type and values of its fields.
- (45) A library is a package of definitions of artefact types, such as agents, services, data sources, and domain artefacts. Partner developers may build libraries, while model developers may use these built libraries to construct models.
- (46) Services as referred to herein may be APIs to an external application which may be used within a model. There is a decorator which defines a service within a library. Services may provide actions that can be performed.
- (47) The term 'service interface' may refer to an API to an external application which may be implemented by a service. This may create an abstraction between an action and the implementation.
- (48) Agents may be software that take on roles, or implement actions. Agents may act via services to effect the role using user credentials for which the user has granted permissions. An agent may undertake their role by automatically performing one or more task. Libraries may not define tasks that an agent is to perform in order to undertake their role. Libraries may define roles, and agents configured to undertake those roles may not be constrained to a particular methodology or predefined set of tasks within that role.
- (49) Templates refer to blueprints of a domain or model. Using a template may speed up the configuration and definition of a model.
- (50) An SDK is a collection of code and utilities used to facilitate building models using the platform **110**. SDKs may operate using particular programming languages, e.g., python.
- (51) The term simulation may refer to the application of values to a subset of the elements of a business mechanic, in order to get the values for other elements via relationships defined in the mechanic.
- (52) FIG. 2 shows an example of a system comprising a decision infrastructure **200** and connected components configured for identifying and implementing actions autonomously based on a digital twin model of an entity.
- (53) The system comprises a user interface component **201**, e.g., for viewing and managing a model of an entity. A console **203** is provided, and may provide repositories for agents. Agents **205** may connect to the decision infrastructure **200** to access models in which they have been engaged to perform a role. The agents may access user credentials to perform their role using those user credentials. Services **207** also connect to the decision infrastructure for providing functionality. The agents **205** and services **207** may need to be registered at the console **203** before being implemented and engaged by the model. The agents **205** may perform their respective roles by accessing services **205** using the user credentials.
- (54) Entity data, e.g. business data, **209** is also accessible via the decision infrastructure.
- (55) The decision infrastructure **200** may comprise tools **211**, such as advanced reasoning tools which may leverage AI or large language models (LLM), continuous forecasting, simulations and problem solving tools, communication functionalities, agent infrastructures for interfacing with agents, the shared model of an entity which is accessible to partners, services and agents, and data access tools.
- (56) FIG. 3 shows a schematic diagram of an example sandbox environment and agent providers connecting to the environment.
- (57) FIG. 3 shows a block **310** representing a first agent provider, provider A. FIG. 3 also shows a block **320** representing a second agent provider, provider B. each agent provider **310**, **320** runs an agent on their own infrastructure.
- (58) The agents communicate via a platform API **340** for the platform providing the sandbox

environment.

(59) FIG. 3 also shows a domain specific application **330** accessing the platform via the API **340**.

(60) A customer-specific UI sandbox environment is represented by block **350**. A plurality of blocks **350** are shown overlapping each other to represent that the platform supports plural customers. The customer environment **350** comprises a model environment **360** for constructing a model **370** of an entity. Example artefacts are represented by interlinked circles and triangles within the model **370**. The triangles in the model **370** represent roles. Agents may be engaged to fulfil a role, as indicated by the connection between agent provider B **320** and a first triangle **371** representing a role, via API **340**. Similarly, agent provider A is shown to be linked via the API **340** to a second triangle, and the domain specific application is linked via the API **340** to a third role/action artefact in the model **370**.

(61) The platform is associated with an artefact library **392** which provides available artefacts based on a current state of the model. The available artefacts may be provided based on inspection by the runtime engine of a loaded library.

(62) FIG. 3 further shows an agent registry **394** comprising agents which have been registered, and an agent admin dashboard **396**.

(63) Reference numeral **380** denotes a data store which may be accessible via the API **340** to an agent provider (e.g., **310**) to assist in performing the role the agent has been engaged to perform.

(64) An example model builder user interface is described with reference to FIG. 1. On entering the model builder, libraries may be installed. A library may contain a plurality of template artefacts (e.g. artefact classes in the OOP sense) and predefined links between artefacts. On installing a library, a set of available artefacts is provided via a user interface (UI). A user may insert an artefact into the model based on the set of available artefacts.

(65) A runtime engine may be configured to load libraries and to interpret the libraries to identify the set of available artefacts. That is, the runtime engine may inspect a library to assess which artefacts are suitable for inserting into the model, and may provide the set of available artefacts on the UI for inserting into a model.

(66) A model of an entity may be built using one or more available artefacts and by linking artefacts. The underlying runtime engine may reason about connections between artefacts based on the installed library. The runtime engine inspects the libraries that are installed and the current state of the model to offer further available artefacts which may be linked to existing artefacts. Dynamic context matching techniques described later herein provide the technical framework enabling the runtime engine to identify contexts, update the model, and instruct external agents to act on behalf of the entity.

(67) The user interface may provide a sandbox environment supported by the runtime engine. A model may be created within the user interface by a user, by selecting available artefacts and implementing the artefact in the model.

(68) In one example, the user may initially create an organization artefact representing a real-world organization. The organization artefact is an instance of a type of data structure, including metadata. The organization artefact may represent a first artefact in the model. By including an organization artefact in the model, the runtime engine may identify a set of further artefacts which are available in view of the model having an organization, and in view of the links defined in the library.

(69) In another example, the library may provide a predefined template for an entity. The template model may comprise a plurality of artefacts and links therebetween to represent respective components and links of a real-world entity.

(70) The entity being modelled may be a business linked to an organization. The model may represent the organization as a first artefact and represent the business as a second artefact. The business may be linked to the organization by a first type of link, e.g., a 'business_of' link indicating that the business is a business of the organization.

- (71) Artefacts may also be added to represent real-world activities of the business, real-world metrics, real world budgets, objectives and other aspects of the entity. The model may include respective links between these artefacts to indicate their real-world association with the business.
- (72) The metric artefacts may represent information that is important to the business. Examples include revenue and profitability.
- (73) Objective artefacts may be linked to a metric artefact to indicate a goal or aim in respect of the associated metric. For example, a target revenue may be represented as an objective artefact linked to the metric artefact, where the metric artefact is then linked to the business artefact.
- (74) Real-world operations of the business may also be represented by artefacts in the model. For example, the business may engage in online retail or eCommerce activities. An artefact may be included in the model to represent the real-world eCommerce operation, and linked to the business within the model.
- (75) The eCommerce operations of the business may include one or more channel, e.g., an online blog and an online shop. Each channel may be represented by a respective artefact and linked to the business. The blog channel may be connected in the model to a blog artefact representing the blog itself. Metric artefacts for performance of the blog, such as viewing figures and reach etc. may be configured in the model. Objective artefacts for the blog metrics may be configured too, defining objectives related to the measurements made in respect of the blog.
- (76) For the online shop channel, a shop artefact may be configured in the model to represent the shop, with one or more related metric artefact and objective artefact representing the real-world metrics by which the shop channel performance is measured, and defining goals for the metrics.
- (77) Fundamentally, the model represents a digital twin of a business or other entity. In the example of modelling a business, artefacts may be included in the model to represent various aspects of the business, business goals, and revenue channels, and to represent real world metrics for each aspect including target/objective for the metrics. Business mechanics/physics may also be encoded as artefacts in a model. Parts of a business such as operations, finance, marketing, or other departments may add artefacts representing business mechanics. The examples herein are not limited to revenue generation.
- (78) Other examples of channels, metrics and objectives than those discussed above may be configured. The principles above also extend to other sectors and problem domains than eCommerce.
- (79) The model may further include artefacts that define a role within the business. That is, an artefact may represent a role for taking an action within the business. In one example, an artefact representing a role of generating blog posts may be included in the model and associated with the blog artefact. As discussed above, a role artefact may be provided within the set of available artefacts in the UI based on the runtime engine's interpretation of the library, where the library defines role artefacts and identifies the association of particular roles with particular artefacts. For example, the blog generation artefact may be provided within a set of available artefacts based on inclusion of a blog artefact within the model, and based on the structure of the library.
- (80) An agent configured to perform a role may be engaged via the model. At runtime, an agent may take real-world actions on behalf of the business based on the digital twin model of the business. However, agents themselves and the code they run may not be defined in the library. Agents may be third party agents suitable for implementing a particular role/action. Ability to perform an action in certain contexts of artefacts in the model, i.e., the interface of the roles and artefacts, are defined in the library.
- (81) Third party agents may offer practical delivery of a role defined in a library. The user interface may enable a user to select an agent and engage that agent to perform a role defined in the model.
- (82) In some examples, a plurality of agents may be able to provide a same role. The role is represented in the model, and a user may arbitrarily select one of the available agents to engage, for that agent to practically deliver the associated role.

- (83) An agent may act on behalf of the business by performing a role it was engaged to undertake. The model may act as a smart ecosystem. Examples herein make the ecosystem (as the business) intelligible for the AI agents so that they can interpret the given service in relation to the common business model, and have the capability, when possible, to even estimate and measure the impact of that service on a business metric or objective.
- (84) The model building tool may be hosted on a platform with which organizations register to build models. One or more libraries may also be registered with the platform. Registered libraries may then be available for loading in the runtime engine to allow models to be constructed based on the library.
- (85) Services may also be registered with the platform to provide functionality. For example analytics services and links to web services may be registered enabling those services to be represented in the model. Services are available to make real-world decisions and take real-world actions. That is, agents may operate through a service to perform their associated role. In one example, a blog-posting agent configured to upload content to a blog may be engaged to satisfy a role, where that role is encoded as an artefact within the model. The agent may act through a registered service to deliver its role in the real-world—i.e., posting real-world blog content.
- (86) Agents are also registered with the platform. Registered agents may then be selected in the model building tool and linked to an associated role to enable actions associated with the role to be delivered by that agent at runtime. Multiple agents may be registered in respect of a single role, and a user, when engaging an agent to fulfil a role in their model, may select from a set of suitable agents.
- (87) Similarly, a single registered agent may be capable of fulfilling a plurality of roles. In such cases, a same agent may be recommended as a suitable agent in respect of a plurality of different roles within the model. The user may select to engage the same agent for multiple different roles, where those roles are represented by distinct artefacts in the model.
- (88) Agents may operate using credentials of the user. The model may comprise user credentials for a service. An agent seeking to deliver its assigned role through a service may interface with the model to retrieve credentials of the user, and then the service may be engaged by the agent using the retrieved credentials.
- (89) Permissions may be required to allow an agent to access user credentials via the model. However, the user may provide permission to the platform, not each agent individually. This allows a single grant of permission to enable multiple registered agents to interface with the platform and model to retrieve and use the user credentials when delivering their role.
- (90) An agent may operate proxy to the decision infrastructure or may operate directly with the service. However, the user does not grant permission to individual agents. The user grants permission to the platform hosting the model building tool, and any registered agent that is engaged to fulfil a role is able to access the user credentials based on the permission granted to the platform.
- (91) One of the challenges of implementing agents to work with services belonging to third-parties is credentials management. Current service authentication techniques have evolved from the principle of a client providing access directly to an application/service to retrieve user data or invoke actions. In the situation of having multiple external agents hosted remotely and separately interacting with services, known techniques may require users to provide credentials to every agent, every time the agent wants to gain access or use a service. This quickly becomes a problem, as most users may not have the organizational permissions to do this, or the knowledge how to do it, let alone manage the geometrically increasing number of credentials issued.
- (92) The present system may contain a mechanism whereby an organization can define the credentials once for a given service, which can then be used by agents via the service-type artefact instances defined in the model. These are instantiated via the appropriate pre-configured credentials. This way, one organizational administrator can configure the credentials, but many actants can use them (without direct access) and external agents can use instantiated services.

(93) To facilitate this, the system offers an API proxying service which injects the credentials into service calls, without revealing the credentials to the external agent. Furthermore, this offers the opportunity to add model-specific limitations over and above the access security of the service itself. For example, whilst the service might give access to all the data the API proxy might be configured to filter out PII (personally identifiable information) data.

(94) The platform represents a meeting point and mediator between external agents, available services, and organizations using the modelling tool.

(95) The platform (e.g., the decision infrastructure) that hosts the model building tool acts as a wrapper or interface between the agents, models and services.

(96) More generally, a service may be understood as a function which can be interrogated via an API. For example, external analytics services, services, and internal company data may all be accessible and interrogable via an API and may therefore be registered as a service. An agent may then be engaged to leverage the service to act on behalf of the entity being modelled.

(97) Roles such as blog posting are defined as an interface in a library. Corresponding services are configured to implement that interface, and agents are engaged to operate that interface and perform the role through the service based on the role defined in the library.

(98) During model building, an initial model may be constructed with various artefacts and links therebetween, including artefacts representing roles. Agents may then be engaged to fulfil roles, e.g., via services.

(99) In some examples, engaged agents may recommend actions that they can take to fulfil their role. Recommended actions may be implemented within the model as temporary artefacts representing outputs of the action. The temporary artefacts may be visible to other agents providing different actions, who may in turn provide temporary artefacts identifying actions that may be taken based on the model and other temporary artefacts.

(100) Agents may undertake one or more task autonomously as they undertake the role they are engaged for. For example, an agent configured to undertake a role of writing and posting blog articles may automatically undertake the tasks of identifying an article theme, generating content, formatting the content, and otherwise preparing the article for posting. A temporary artefact may be surfaced in the model indicating that an article is available to post. A user may be able to accept or reject the recommendation to post an article. In other examples, the agent may autonomously post the article and may instantiate an artefact in the model, representing the article within the blog.

(101) Whilst building a model, artefacts and links may be used to accurately recreate the current state of the entity being modelled. When agents are engaged and the model is live, i.e., at runtime, agents may perform their role to produce outputs, and automatically cause a new artefact to be included in the model to represent the output.

(102) Temporary artefacts may be understood as agent proposal artefacts which may be accepted by a user to add that artefact to the model. An agent may propose to include a new artefact in the model because a particular combination of artefacts with certain properties exists in the model, and the agent recognizes that its role is available when that particular combination of artefacts is present.

(103) The models described herein may be seen as dynamic models since artefacts may be added or removed in view of actions undertaken by agents.

(104) Situations where an artefact can be added to the model based on other artefacts and roles in the model may be captured as rules in a state machine. States in the state machine are associated with respective available agent engagements and transitions between states are defined by the entity model. An update to the model may cause a transition in the state machine, which in turn causes a new set of agent engagements (associated with the new state) to become available.

(105) A model may further encode mechanics of the entity. For example, relationships between metrics may be encoded in the model, based on real-world metrics and real-world relationships between those metrics. Metrics are represented as metric artefacts, with dependencies defined by

mathematical equations, analogous to a physics model.

(106) An entity model provides the possibility to simulate the entity and enables forecasting based on the digital twin of the entity.

(107) As discussed above, the models are supported by an execution or runtime engine which can implement a library to build models of an entity. The runtime engine inspects what is available within the library based on the current form of the model and provides a set of available artefacts and roles etc. The runtime engine acts to observe the state of the machine and the available libraries that have been imported and provides the set of possible actions.

(108) As the state of the model changes, as artefacts are added or removed, the set of available actions is also updated based on inspection of the libraries and the model by the runtime engine.

(109) Reference is made to FIG. 4, which shows an example representation of a model, including artefacts representing an organization **410**, and an associated business **420**. The model further includes a budget artefact **430** representing a budget of the business, and two metrics of the business **440a**, **440b**. The first metric **440a** is associated with an objective artefact **450** defining an objective in respect of the first metric **440a**.

(110) The second metric artefact **440b** represents a profitability of the business.

(111) The business artefact is also linked to an operation artefact **460** which represents an operation, e.g., sector of operation of the business. Two channel artefacts **470a** and **470b** respectively represent channels of operation of the business. The first channel artefact **470a** represents a blog channel, and the second channel artefact **470b** represents an online shop.

(112) The blog and online shop are represented by respective artefacts **480a** and **480b**.

(113) The blog artefact **480a** is associated with a blog metric artefact **440c** which is a metric artefact associated with the blog. The metric may relate, for example, to blog engagement etc.

(114) A blog generating role is also represented by artefact **490**. The blog generating role represents actions that may be taken by an agent to perform a role such as generating content for a blog and, for example, posting that content on the blog.

(115) An agent may be engaged to perform the role represented by the role artefact **490**, as discussed above.

(116) The artefacts in FIG. 4 are linked together to indicate their interdependence and association with one another.

(117) The description that follows provides, with reference to FIGS. 7a-7d an example flow in which an initial model is updated automatically when an agent engaged to perform a role initiates a recommended action.

(118) The flow begins with an initial set of artefacts, and links between artefacts. The initial model may be a template, e.g. a selected template from a template registry. The initial model may be based on a template from a template registry and subsequently edited by a user to provide the initial model. In another example, the model may be user defined by creating artefacts individually. FIG. 7a represents the artefact structure of an initial model, constructed based on a template model. Other than a business artefact representing an example shirt business, and an example eCommerce operation artefact, FIG. 7a does not give specific examples of other artefacts in the model. The various unlabeled artefacts may represent an revenue channels, roles (agent roles), metrics, websites or any other concept of the entity being modelled. The business artefact and eCommerce Operation artefact are emboldened for clarity in FIG. 7a.

(119) The runtime layer may inspect a library against the initial model to determine available options for expanding the initial model. The available options may be presented on a user interface for selection by a user. The available options may include options to run recommended agents, e.g., role artefacts. FIG. 7b represents an exemplary UI feature comprising a list **720** of available options for expanding the model. In FIG. 7b, an option bar **710** is provided for the user to navigate the user interface. A 'create' option is selected in the option bar **710**, and accordingly the list **720** of available options for creating an artefact is displayed. FIG. 7b shows a user cursor hovering over a

'New Email Channel' option in the list **720**.

(120) FIG. **7c** shows an example UI feature comprising an option to engage a recommendation agent, e.g., to fulfil a role represented in the model. In FIG. **7b**, the engage tab is selected in the option bar **710**, and a single engagement option **721** is presented for selection in the available selections block **720**.

(121) Selecting the option to engage the recommendation agent runs the recommendation agent. Running the recommendation agent causes action recommendations to be added to the model. FIG. **7d** shows the same model as in FIG. **7a**, wherein the model further includes a role artefact (or agent artefact) for an eCommerce consultant (artefact labelled Y.sub.1) and a plurality of action recommendation artefacts (labelled X.sub.1, X.sub.2, X.sub.3). The action recommendation artefacts and agent artefact are shown with dashed borders and with dashed links between them and other artefacts, to distinguish the model state of FIG. **7d** from that of the initial model in FIG. **7a**.

(122) A recommended action may then be initiated. Initiating a recommended action comprises calling an external agent. The external agent that is called interacts with an external service, or API for an external service, via the platform hosting the model (e.g., via Honu). An automation action is performed by the interaction of the external agent with the external service.

(123) The model is updated to include one or more artefact, indicating that the automation action has been performed.

(124) The runtime engine then inspects the updated model against the library to update the available options.

(125) In the above examples, a graphical user interface (GUI) is provided to visualize the entity model as it is created and updated. Whilst useful, such a GUI is not essential, and the above description pertaining to the visualization of the entity model applies more generally to the underlying model itself, whether or not the model is graphically represented.

(126) Dynamic Context Matching

(127) Dynamic context matching refers to a process in which the runtime engine identifies structures of artefact relationships ('contexts') in an entity model, and responsively enables behaviors to be made available and/or generates an instruction for an action to be performed (e.g., by an autonomous agent).

(128) The inventors have built an effective mechanism for decision-making about the structure and operation of an entity such as a business, using specialized technology-based agents communicating and collaborating through a common model of the entity. The model is connected to digital services such that state, action, and feedback can be used to affect the operation and management of the entity in a wholly-automated and potentially autonomous manner. The system also provides mechanisms allowing human intervention to guide or make decisions alongside the agents.

(129) Reference is made to FIG. **8**, which shows an example system **800** in communication with an external agent. The system is configured to enable automation of an entity such as a business, by modelling aspects of the entity and using the model to inform decisions, and identify contexts in which certain actions can potentially be automated and exploited by actants using the model.

(130) The system **800** of FIG. **8** comprises a runtime engine **810**, an entity model **820**, and a library **830**. The entity model **820** is dynamic and evolves to reflect real-world changes and contexts of a real-world entity being modelled. The entity model **820** encodes structural information of the entity being modelled, and further encodes processing cues for informing runtime engine actions in view of the model. The library **830** holds template or candidate model features (e.g., artefacts) which may be instantiated in the entity model when that feature becomes relevant to the real-world entity.

(131) The runtime engine **810** is executed on one or more processor **812** coupled to computer memory **814**, which holds code of the runtime engine **810** for execution. The runtime engine is configured to perform processes within the central system **800**, such as inspecting entity models **820**, accessing libraries **830**, and transmitting instructions to external agents **860** via interfaces **850**

between the system **800** and the external agents **860**. Processes in the runtime engine **810** may be performed by the processor **812** based on computer-executable instructions held in the memory **814**.

(132) The entity model **820** provides an explicit model of the structure of an entity, and further provides information to inform the runtime engine of how to act based on the structures in the model. The example entity model **820** of FIG. **8** comprises an information model **824** and an execution layer **822**. The information model **824** and execution layer **822** represent different aspects of the overall entity model **820**. The information model **824** represents a structural model of the entity, and the execution layer **824** of the entity model **820** represents processing cues that inform the runtime engine **810** on how to interact with the model when automating actions and autonomously updating the model **820**.

(133) Instances of data structures referred to herein as ‘artefacts’ are provided in the entity model **820**. An instance of an artefact in the model **820** may comprise structural information defining a concept in the domain of the entity being modelled, and further defining the relationship of that concept to other artefacts in the model. An instance of an artefact may also be configured as a particular ‘type’ of artefact. The artefact type may indicate an execution cue that informs the runtime engine **810** on how to interpret the artefact in the model. In FIG. **8**, the information model **824** represents the structural information of artefact instances in the model **820**. The execution layer **822** represents type information of artefact instances in the model **820**, which gives execution cues to the runtime engine.

(134) By way of example only, object-oriented programming techniques such as classes and objects may be leveraged to encode structural information in artefacts. Relationships between artefacts in the domain of an entity being modelled, as represented in the information model **824**, may also be encoded based on the class structure. An artefact instance may have a particular class defining the concept it represents. Candidate artefacts in the library **830** are configured (e.g., using class structures in object-oriented programming) such that they convey structural information in the model when they are instantiated in, i.e., added to, the entity model. It will be understood, however, that the techniques described herein are not limited to a specific syntax, or indeed to an OOP-based approach (e.g., as noted, an entity model might instead be realized as a graph database or any other suitable data structure).

(135) An artefact instance may be configured with a particular execution cue by being assigned a ‘type’. In some examples, the type of each artefact may be defined using decorators, which are a feature of python code. The use of decorators enables functionality to be applied to the artefacts, without requiring the class definitions of those artefacts to be modified.

(136) An artefact instance in the model may comprise a class object defining structural information, and may further comprise a decorated type that informs the runtime engine of how to process the artefact in the context of the model. Examples of artefact classes and types are provided later herein.

(137) The information model **824** of FIG. **8** shows the structure of an example entity, as defined by structural information (e.g., class objects) of the artefacts in the entity model **820**.

(138) Byway of example, FIG. **8** shows a plurality of artefacts **826**, and an agent artefact (or role artefact) **828**. Each artefact **826** represents an instance of a concept in the domain of the entity. The agent artefact **828** represents a role associated with the entity. Links between the artefacts **826** represent structural relationships between the concepts and roles. In examples where a business is being modelled, the artefacts **826** may represent concepts related to a business domain, and the agent artefact **828** represents a role within the business, e.g., a capability to perform actions on behalf of the business. Similarly, links between artefacts in the information model **824** represent structural links between concepts in the business.

(139) The agent artefact **828** may have a different type (e.g., different decorator) to the other artefacts **826** in the entity model **820**, so that the runtime engine interprets the agent differently to

other artefacts **826** which do not represent agents. However, at the information model level, it is the structure and relationship of the artefacts **826**, **828** that is represented, and this structure may be encoded using instances of different classes. Information encoded by artefact types is not represented in the information model **824** of FIG. **8**.

(140) Each of the artefacts **826** may represent a different aspect of the entity being modeled. Any conceivable feature of an entity may be modelled and represented by an artefact data structure defining that feature. As discussed later herein, libraries of template artefact data structures may be compiled and may be accessible to the runtime engine **810** for inclusion in the entity model **820**. For instance, when a concept for which a corresponding artefact has been defined becomes relevant to an entity being modelled, an entity model **820** for that entity may be updated to include an instance of the corresponding artefact.

(141) The execution layer **822** augments the information model **824** with behavioral cues that tell the runtime engine how to process the information model **824**. The runtime engine **810** may instruct external agents to perform actions on behalf of the entity based on the information model **824**, as informed by the execution layer **822**. In practice, each artefact instance in the entity model **820** may be defined with a particular class for conveying structural information, and with a type for conveying execution cues to the runtime engine.

(142) The runtime engine **810** is shown to be connected to the entity model **820**. The connection between the runtime engine **810** and the entity model **820** represents the ability of the runtime engine **810** to inspect, modify, interpret, and otherwise interact with the entity model **820**. Examples of ways in which the runtime engine **810** interacts with the entity model **820** are described elsewhere herein.

(143) The system **800** further comprises one or more library **830**. The library **830** comprises a plurality of candidate artefacts **832**. Candidate artefacts **832** are template artefact data structures that may be introduced to an entity model **820** (as an instance of that candidate artefact) if a concept represented by the candidate artefact **832** becomes relevant to the entity being modelled. Actions of external agents, for example, may cause a concept to become relevant to the entity being modelled, and may then be represented by a corresponding artefact instance. Various candidate artefacts **832** may be defined in a library, and the underlying template data structures may similarly vary depending on the feature the candidate artefact **832** is configured to represent.

(144) The library **830** further defines contexts **834**, which are predefined structures of artefact relationships within the entity model **820**. The contexts may define states in the information model **824**, and may require particular structures within the entity. The contexts may define structures of artefact relationships independently of the execution cues of each artefact. Whilst artefacts represent real-world concepts pertaining to an entity, contexts defining structures of artefacts may represent real-world situations or states of the entity.

(145) In FIG. **8**, block **834** represents a context defined in a library, and block **836** represents a corresponding predefined structure of artefact relationships. Dynamic context matching refers to a process by which the runtime engine is configured to identify predefined structures of artefacts (contexts) in the library **830**, identify matching structures of artefacts in the entity model **820**, and responsively update the model **820** to indicate that the model includes the corresponding context **834**. That is, when the predefined structure **836** is present in the entity model **820** (e.g., as a substructure of artefacts **826**, **828** in the model **820**), an instance of the context **834** is added to the model. That is, the model is updated to indicate the context is present in the model.

(146) Situations where a context **834** can be added to the entity model **820** based on a structure of attributes **836** may be captured as rules in a state machine. That is, the predefined artefact structure **836** of a context **834** may be associated with a state in the state machine, and identifying a given state may cause an update to the model **820** by the runtime engine **810**, to include an instance of the context.

(147) In response to updating the model to include an instance of the context, the runtime engine

810 may be configured to generate an instruction for an autonomous agent to perform an action.

(148) In this way, entities can be modelled accurately, and the model can be updated autonomously as the entity evolves. Further, by enabling interface with external agents to perform real-world actions, the entity itself can be made autonomous and can perform actions based on the entity model **820**, as informed by contexts **834** in the library **830**.

(149) The present disclosure provides the technical foundations for an autonomous entity to make decisions and perform actions, and to evolve and adjust the way the entity operates based on the model. Libraries **830** can inform the decision making and evolution of the autonomous entity. In the example of an entity being a business, libraries **830**, candidate artefacts **832** and contexts **834** may be configured such that entity models **820** evolve according to particular business models, e.g., with particular tolerances for risk, or with a focus on particular revenue channels. Aspects of the present disclosure, however, relates to the technical foundations for enabling autonomy of entities, enabling dynamic updating of entity models and instructing agent actions by identifying contexts in models.

(150) The entity model **820** and libraries **830** may be held in computer storage of the system **800**, and the runtime engine **810** is configured to access the computer storage to access the entity model **820** and the libraries **830**.

(151) System **800** further comprises one or more interface **850**. The interface may be a suitable programming interface such as an application programming interface (API) configured to enable the system to transmit instructions to external services and actors. The system **800** is configured to centrally coordinate the actions of the entity, but may interface (e.g., via interface **850**) with external agents **860** to instruct the agent **860** to perform the action on behalf of the entity.

(152) The interface **850** enables entities using it to discover what actions are possible to be applied, including adding and removing artefacts, engaging agents, making calls to configured services via proxied endpoints, and conversing with humans or agents engaged with the model.

(153) There may be an additional administration API through which entities register and manage agents, libraries, services, and service credentials. The principle of developing API-first enables all the capabilities to be equally available to humans as to agents—this means that anything a human might do (e.g., create a library, develop an agent, register an agent, engage an agent) could also be done by an agent with sufficient permissions and abilities.

(154) The external agent **860** may be an autonomous agent and may be configured to perform an action on behalf of the entity being modelled or engage a service on behalf of the entity being modelled. External agents may be hosted on remote servers separate from the central system **800**. External agents **860** may communicate with the model via a bi-directional secure protocol implemented through language specific modules on top of a common REST API.

(155) Particular contexts in the library may be associated with particular actions to be performed when that context is identified. The runtime engine **810** may be configured, in response to identifying a context, to cause the associated action to be performed. On identifying the context, the engine **810** may determine an action to be performed based on the context, and generate an instruction for an external agent to perform the action. The instruction may be transmitted to the external agent via an interface **850**.

(156) FIG. 9 shows a flowchart representing an example dynamic context matching process. At a high level, the process of FIG. 9 involves the runtime engine inspecting the runtime model, assessing whether the structure of the model matches any predefined artefact structures (contexts) in the library, assessing whether any existing context instances in the model are no longer satisfied, updating the model to add and/or remove context instances, and instructing an action to be performed. For the purpose of the description herein, a ‘satisfied context’ refers to a predefined structure of artefacts which is exhibited in an entity model. That is, for a particular context in a library to be a satisfied context, the entity model **820** must exhibit the structure of artefact relationships defined by that context in the library. A context instance is a representation of a

satisfied context in the entity model.

(157) FIG. **9** begins at step **S901**, wherein a trigger event occurs. The trigger event is an event that causes the runtime engine to search a change in context within the entity model. One example of a trigger event is an update to the model. Referring to the example of FIG. **7b**, such an update occurs when one of the available options **720** is selected and this selection triggers the addition, removal or modification of an artefact. This, in turn, may cause a new context to be satisfied or could mean that a context that was previously satisfies is no longer satisfied (either of which can result in a change in the set of available options). Another example of a trigger event is an update to a library, or the addition of a new library. For instance, the updated or new library may define a new context.

(158) At a next step **S903**, in response to the trigger event, the runtime engine performs a dynamic context matching process. The dynamic context matching process **S903** comprises assessing the structure of artefact instances in an entity model and determining, based on a library of predefined contexts, which contexts are satisfied, and which are not. The model is then updated to add context instances for satisfied contexts and to remove context instances for contexts that are no longer satisfied.

(159) As part of the context matching step **S903**, the flow comprises steps **S905-S909**. At step **S905**, contexts in the library which are satisfied by the entity model are identified.

(160) At step **S907**, context instances in the entity model which are no longer satisfied (because the model does not exhibit a structure of artefacts defined by the corresponding contexts in the library) are identified.

(161) At step **S909**, the entity model is updated to add and remove context instances based on steps **S905** and **S907**. The runtime engine updates the model to add context instances corresponding to contexts which are identified as being satisfied at step **S905**. The runtime engine removes context instances for contexts that are no longer satisfied, as identified at step **S907**.

(162) At a next step **S911**, an instruction is generated for causing an external agent to perform an action. For example, the runtime engine may generate an instruction for an external agent to perform an action on behalf of the entity being modelled. As discussed elsewhere herein, an external agent may be engaged to perform a role for the entity, where the role may be represented by an instance of an agent artefact in the entity model.

(163) Reference is made now to FIG. **10**, which shows a block diagram representing an action being instructed in response to a context matching process. FIG. **10** illustrates an example set of artefact relationships which may define a context, and shows how an action may be instructed by the runtime engine in response to identifying the context in the model.

(164) FIG. **10** shows an example entity model in a state **1000a**. It will be understood that the links between artefacts in the model represent structural relationships of the entity. The types of each artefact, which provide the execution layer **122** of an entity model, are not represented in FIG. **10**.

(165) The model, in state **1000a**, comprises an agent artefact **1004**, a plurality of other artefacts **1002**, and a new artefact **1008**. Particular reference is made to artefact **1** (**1002a**), artefact **2** (**1002b**), artefact **3** (**1002c**), and new artefact **1008**, which are relevant later in the description of FIG. **10** when a context is identified in the model.

(166) The agent artefact **1004** represents a role **1006**, or a capability for particular actions to be carried out on behalf of the entity being modelled.

(167) In the example of Figure, a trigger event **1030** initiates a context matching process. In FIG. **10**, the trigger event comprises an update to the model to introduce a new artefact instance **1008** to the model. Model state **1000a** represents the model after the trigger event, wherein the model comprises new artefact instance **1008**.

(168) Block **1010** represents a context matching process, e.g., step **S903** in the example of FIG. **9**. Block **1020** represents the runtime engine performing the context matching process. The runtime engine, as informed by contexts defined in the library, identifies a context which is satisfied by the model in state **1000a**. The context matching process is initiated by the trigger event **1030**, in which

new artefact **1008** is introduced to the model.

(169) In the example of FIG. **10**, a context in the library is identified as being satisfied by the model in state **1000a**. For example, by updating the model to include new artefact **1008** based on the trigger event **1030**, the model now satisfies a context defined in the library. In response to identifying the satisfied context, an instance of the satisfied context is added to the model, resulting in an updated model in state **1000b**.

(170) In model state **1000b**, artefacts **1-3** (**1002a-c** respectively), the agent artefact **1004**, the new artefact **1008**, and interconnections between the artefacts, are represented with double-stroke lines. The context identified as being satisfied in the dynamic context matching process **1010** has an associated structure of artefact relationships. The double stroke lines have been applied to artefacts and interconnections which exhibit this structure of artefact relationships in the model. That is, the double stroke lines in FIG. **10** represent the context instance being added to the model. The model in state **1000b** includes a context instance corresponding to the satisfied context identified at block **1010**.

(171) As discussed previously, a context may be associated with an action to be performed upon identifying the context in a model. The runtime engine is configured to generate an instruction for an external agent to perform the action, in response to identifying the satisfied context. Block **1040** of FIG. **10** shows the external agent being instructed in this way.

(172) The external agent may be engaged to perform actions via techniques described elsewhere herein. The agent artefact in the model abstractly represents a real-world role within a real-world entity that is being modelled. An external agent may be engaged to perform actions to fulfil this role in the real-world.

(173) Block **1050** shows a simplified representation of the predefined structure of artefact relationships associated with the satisfied context in this example. The satisfied context in the example of FIG. **10** requires a structure of artefact relationships whereby the new artefact **1008** is related to artefact **1** (**1002a**), artefact **1** (**1002a**) is related to artefact **2** (**1002b**), artefact **2** (**1002b**) is related to artefact **3** (**1002c**), and artefact **3** (**1002c**) is related to the agent artefact (**1004**).

(174) Examples of artefact types, i.e., components of artefacts defining execution cues for the runtime engine, are now described. The examples below are not exhaustive, and other artefact types may be configured to apply functionality to artefacts in a model.

(175) A first example is a generic artefact type. The generic artefact type may define passive entities and concepts relevant to a problem domain. The generic artefact type may be applied to artefacts representing an organization, a business, a contract, a business proposal etc. This example list of passive concepts is non-exhaustive, and many other passive concepts in a problem domain may be provided with a generic artefact type in an entity model.

(176) An agent type may be applied to artefacts representing roles in the entity, i.e., capabilities to perform actions. Agent type artefacts may configure the runtime engine to allow real-world external agents to be engaged to act to perform a role defined in the artefact. Engaged external agents may interface with, read, and change the structure of the model. Agent type artefacts are active elements within the model and can originate activity in the system. External agents engaged to perform the roles encapsulate domain knowledge, decision-making, and problem solving logic, and execute remotely, interfacing with the central system. An example of an agent-type artefact is a pricing agent, which may represent a role of determining prices for products sold by an entity. An external agent engaged to undertake the role of the pricing agent may, in response to instructions from the runtime engine when a relevant dynamic context is identified, adjust a price of a product on an online store, for example. The external agent may interact with the model to identify and retrieve relevant data for performing its role.

(177) A context type defines a situation, condition, or state of the model that when satisfied enables dynamic and context-aware behavior, such as the opportunity to engage a specific external agent, retrieve data pertaining to a particular metric, or be in the position to publish a social media

message.

(178) A metric type may be applied to artefacts defining a measurable value or performance indicator of interest within the problem domain of the entity being modelled. In the context of modelling a business, examples of metrics may include revenue, number of sales, click-through rate, sale conversion rate, and other examples of measurable values.

(179) An objective type may be applied to artefacts defining goals or desired outcomes that need to be achieved within the business to progress its state or structure. The objective type, when applied to an artefact in the model, may express the need for an external agent to be engaged to take action towards satisfying the objective. Artefacts of the objective type may be instantiated by human users or by external agents.

(180) A context instance in a model may identify other artefacts in the model to which an objective type and an agent type have been applied. The context instance, which may dynamically be instantiated in the model based on dynamic context matching techniques described herein, thereby identifies a particular objective of the entity and a role for acting to handle that objective. External agents in the real-world are engaged to perform the role to take actions to satisfy the objective. These agent actions may be instructed by the runtime engine in response to identifying the context.

(181) The artefacts and relationships in the information model of the entity, and the artefact types of each entity providing the execution layer, therefore enable the central system to automate real-world actions on behalf of the entity based on the contexts of the model, as well as allow external agents to update the model to reflect real-world changes to the entity.

(182) Another example artefact type is the service type. Artefacts of the service type represent real-world services, such as online shopping platforms, analytics platforms, courier services, etc. Services can have actions invoked upon them to make changes in the real world, e.g., by instructing external agents to act to use real-world services, based on the agent and the service being represented by suitable artefacts in an entity model.

(183) A datasource artefact type may be defined. The datasource type represents when data for a particular metric can be retrieved from a service, and what indexing and aggregation is available in the service itself.

(184) As noted above, artefact types may be encoded using decorators in python. An artefact instance in a model may be encoded with structural information defined by the class, and the class object may be decorated with a particular type to provide the execution cues. In python, an '@' prefix is used to denote a decorator, and an artefact instance may therefore take the form: @type class ArtefactClass: #class definition

(185) FIG. 11 shows a specific example of modelling an organization using a model of explicit artefacts which represent core concepts. FIG. 11 shows how the presently described modelling techniques are more easily extensible than other known tools, requiring no change to base definitions of candidate artefacts. In the example model of FIG. 11, a set of businesses **1121**, **1123**, **1125** are associated with an organization **1110**, and the model is extended to provide business descriptions **1141**, **1143** to some businesses within the model.

(186) Labels on the arrows between the organization and the businesses represent artefact parameters structurally linking the businesses **1121-1125** to the organization **1110**.

(187) Template Organization **1151** and Business **1151** artefacts may be packaged into a first library **1141**. To extend the model to have descriptions attached to the Businesses, most tools would require an update to the candidate business artefact **1151** definition, to add a new parameter name to the Business (e.g. description). However, a user may not have the access to change that base definition. Other tools may suggest creating a new candidate artefact which uses a sub-class of the Business (e.g. BusinessWithDescription). However, it is then a challenge to change all the instances to conform to this new structure.

(188) FIG. 11 shows an alternate solution for allowing extensions to the model. A second library, Library 2 (**1143**) is defined, and comprises a candidate business description artefact **1155**. Instances

of the candidate business description artefact (e.g., instances **1131**, **1133**) may be added to the model and structurally linked to respective business artefacts, without modification to the content or structure of the business artefacts. A link between each business description artefact and its respective business artefact is stored within the business description artefact (rather than the business artefact).

(189) As discussed above, however, behaviors and actions of external agents are not bound to artefact instances themselves, but to patterns of artefacts in the model (contexts). When library **2** (**1143**) is imported, it is possible to define some contextual behavior, i.e. if there is a Business (**1121**) with a BusinessDescription (**1131**) there is an action to get the text of that description. Also, if a Business **1125** doesn't have a BusinessDescription, there may be an action which returns “undescribed”.

(190) To find out what actions are available, the model may be asked for “all actions”, which dynamically changes according to the configuration and state of the model. Built into the approach is the ability to describe the artefacts and actions with parameter-based comments, meaning that it is possible to get a plain English description of the artefacts and actions available. This means that LLM-based agents might use pure natural language approaches to interacting with the model rather than using knowledge directly from knowing the library in use.

(191) Dynamic contexts enable specialization and extension of models without having to alter artefact definitions. Dynamic context matching enables evolution of the model of a business as it grows and changes, and as new capabilities become available as third-parties develop expertise and concepts. Dynamic contexts are the elements which help model processes and practices in a business and represent the state machine of the business—but allow flexibility and extension of those processes and practices, opening them up to additional steps and skills defined in new libraries.

(192) External agents (and optionally humans) collaboratively build the model of the business by importing libraries and invoking actions to build a shared representation of the business. It is a shared concept model about which the agents can then reason. But reasoning is not an isolated activity. Reasoning requires conversation and agreement. The explicit shared model of the business gives a strong foundation for those conversations, which may take place between two or more agents, or between agents and humans.

(193) Whilst the entity model is the primary shared source of communication about the state and structure of the business, as previously mentioned, conversations between actants (e.g., humans and agents) may be enabled to rationalize and agree on changes to the model.

(194) For example, a “BusinessSetUp” agent may ask a number of questions to a human to select the appropriate libraries and build the initial model. An agent might need to seek approval (@objective to approve action) for taking an action from an actant with decision responsibilities. That actant might ask for an explanation of how the requestor came to suggest the action, and might want to provide additional information to change the request (e.g., there may be a limited budget the agent hasn't taken into account)

(195) In another example, an agent may ask a supervisory actant questions to extend the model and provide options and choices which are not yet represented in the actual state of the model (e.g. would you like to optimize for ROI or CashOnHand?).

(196) In another example, a supervisory agent may request all engaged agents to generate proposals for improving revenue generation—this could be done with an @objective and then agents which can engage in an appropriate context with that @objective could ask additional questions to help them generate proposals. There would be one conversation with all the engaged agents meaning they are working from the same information/responses.

(197) With the increasing sophistication of LLMs, the conversational capabilities of the model, along with the plain-English descriptions available for entities and available actions creates an opportunity for expert LLM-based agents to work with a strong model of a business and have

context related conversations with other actants about the operation and structure of the business. (198) The REST API provides an interface for agents to initiate and participate in conversations between humans and agents. Conversations can have many participants and can be linked to a particular context (to provide a strong subject for the conversation). Conversations may be purely agent-based with no humans in the loop.

(199) Conversations such as those in the examples above may be initiated in response to the runtime engine identifying a context in which a decision needs to be made, and responsively instructing an action to initiate the conversation.

(200) The concept of entity mechanics is now discussed. Orthogonal to the elements which are used to model the fundamental structure of an entity and the actants, are some additional elements which define entity mechanics. Entity mechanics define functional relationships between metrics.

(201) Entity mechanics are used when retrieving data and when forecasting or backcasting metrics. It is important that all external agents interacting with a model use the same way of calculating values, otherwise inconsistencies will occur and comparisons cannot be made independently of the agents. For example, there are several ways to calculate weekly averages: Monday-Sunday, Sunday-Saturday, or even weekdays only. These can make significant differences but even more so when thinking about accounting practices and how taxes are allocated, or cost of returns calculated. Entity mechanics being part of the shared model ensures consistency and alleviates certain responsibilities from the agent developer. The platform may provide mechanisms for calculating, solving and forecasting metrics using the mechanics.

(202) Entity mechanics, at their most basic, are expressed as transfer functions between metrics. To take an extremely simple example, $\text{AverageSales} = \text{TotalSales} / \text{NumberOfSales}$.

(203) If the model comprises respective data sources for TotalSales and NumberOfSales, the mechanic function above may be solved to calculate a historic time-series of AverageSales

(204) If the model comprises respective data sources for AverageSales and TotalSales, but not NumberOfSales, a NumberOfSales may be inferred based on the entity mechanic.

(205) If the model comprises respective data sources for AverageSales and NumberOfSales, those data may be statistically characterized, and TotalSales may be forecasted with confidence limits for a period of time.

(206) If the model comprises data sources for all three metrics, all expressed on a weekly basis, an agent could check to see which approach is being used to define the weekly boundaries—and may uncover that the sources are not consistent (implying action needs to be taken to resolve this or correct the data with further business mechanics).

(207) By expressing the functional relationships between metrics explicitly in entity mechanics as first class objects, rather than simple functions, the mechanics may be shared between agents, which ensures consistency.

(208) The combination of the model and the entity mechanics means there is sufficient information to offer full, digital-twin simulations and analysis such that “what if” experimentation can be undertaken to estimate the impacts of tactical and strategic decisions on particular entities, but also would allow agent developers to examine the behavior of their agents under a variety of model conditions.

(209) Entity mechanics are orthogonal to the artefact model but still use the same dynamic context mechanisms to match and select which mechanics to apply. For example, if there is a single data source for Sales and it is expressed in a single currency there is one mechanic required to calculate TotalSales. However, if there is more than one source of Sales and they are in different currencies, a more sophisticated mechanic is required. This can be captured and handled through defining the mechanic with the appropriate context. In either case, an agent can deal with TotalSales as a simple concept and leave the calculations to the platform model and its mechanics.

(210) Reference is now made to FIG. 12, which illustrates a process of instantiating artefacts in an entity model by a runtime engine 1201. In the example of FIG. 12, class object structures and

decorators in Python syntax are used to exemplify the entity modeling and context matching processes. As discussed above, however, the present disclosure is not limited to this syntax and may be implemented in other ways.

(211) In FIG. 12, candidate artefacts **1221-1229** are represented with dashed borders on the right-hand side of FIG. 12. FIG. 12 also shows a plurality of instantiated artefacts **1211-1219**.

(212) From top to bottom, FIG. 12 represents a flow starting from an empty entity model to a model representing an organization with an associated business, where the business has an associated business description. The business description is instantiated based on a dynamic context matching process which causes a descriptive agent to be instructed to generate a description that is included in a business description artefact instantiated in the model.

(213) A first artefact instantiated in the model is an organization artefact **1211**, instantiated based on a candidate artefact **1221** packaged in a library. The candidate organization artefact **1221** includes a parameter defining a name of the organization.

(214) When the organization artefact **1211** is instantiated in the model, a business artefact may then be instantiated, where the business artefact is structurally linked to the organization. Candidate artefact **1223** represents a business artefact, and has parameters defining an organization of which the business is a part.

(215) Block **1203** represents the runtime engine adding a business artefact instance **1213** to the model. By instantiating the business artefact **1213** with a 'business_of' parameter value corresponding to the Organization artefact **1211**, the business artefact **1213** is structurally linked to the organization artefact **1211** in the model.

(216) A candidate agent artefact **1225** for a descriptive agent is provided in the library, and defines an interface for an external agent to provide the corresponding role. The role may be for generating a description of a business.

(217) FIG. 12 shows a description agent artefact **1215** being instantiated in the model.

(218) In the example of FIG. 12, a context **1227** defined in the library defines a predefined artefact structure including a DescriptionAgent artefact, and a Business artefact. That is, when a model comprises both artefacts in the predefined artefact structure of the context **1227**, that context is matched. The context **1227** also defines an 'On_Creation' function which is invoked on instantiation of a corresponding context artefact **1217** in the model. In FIG. 12, the On_Creation function comprises an invocation of an external description agent to generate a description for the business, based on the interface defined in the description agent artefact **1215**.

(219) The runtime engine **1201** matches the model structure with the predefined artefact structure in the context **1228**, as represented by arrows from the business artefact **1213** and description agent artefact **1215** to the runtime engine **1201**. The runtime engine accordingly instantiates a context artefact **1217** in the model. The on_creation function is invoked, and the description agent generates a description.

(220) As a result, a business description artefact **1219** is instantiated in the model, based on a candidate business description artefact **1229** in the library. The candidate business description artefact **1229** comprises parameters for defining a business which is being described (business_of), and for providing a string corresponding to the business description. The instantiated business description artefact **1219** structurally links the business description to the business artefact **1213** by way of the description_of parameter value.

(221) Data mapping is now considered. Artefact instances in entity models represent concepts, and therefore may represent the concept of data pertaining to the entity. The artefacts may only represent the existence of relevant data though, and may not contain the data itself.

(222) However, agents need to access and retrieve data to undertake their roles. The principle in the Data Mapping architecture is that data does not flow through the model, but is accessed directly by the agent. Furthermore, there are different modes of data retrieval depending on the scale and use of the data required.

(223) The combination of the defined metric, service, and datasource artefact types, along with the entity mechanics, may determine the type of datasets that can be retrieved through queries. Datasets can be thought of as arrays of data with columns and row indexes described by metric-type artefacts. For example, a dataset for TotalSales by Date has a single column and a row per date.

(224) Queries may be formulated by agents by reflecting on the metrics available in the model (e.g. TotalSales by Date between 01/01/24 and 31/12/24). Querying is a two-step process: the first step is determining what format and data access methods are available, and the second is to access and retrieve the data. This two-step process may be configured because there are a number of competing demands on accessing data.

(225) Firstly, there is the scale of the data. If an agent wishes to retrieve a large data set for training purposes, it would be inefficient and inappropriate to stream this through a socket connection. It would be better to dump this data into a storage bucket in the cloud, and inform the agent where it was. By contrast, if the agent were only retrieving a single short time-series, returning the data as a language-specific data array might be better. This is where the first step comes into play: determining appropriate methods of access. Once selected, each method of access has its own standard way of retrieval which is the second step.

(226) The runtime engine, which determines how to retrieve data for a particular model configuration, works with the definitions in the model and the entity mechanics in order to convert the abstract query into specific interactions with the services containing the raw data required to calculate the requested dataset. For example, accessing raw data from two separate online store services and combining the data according to an entity mechanic definitions to generate TotalSales by Date.

(227) The majority of this work is undertaken in the processing scope of the agent. Ultimately, the data retrieval calls are routed via an API proxy running in the system, which inserts appropriate credentials and enforces data access retrieval rules.

(228) Library Generation Through Conversation

(229) A further concept that may be applied in the present system is that of library generation through the conversation mechanisms described above. The definitions of artefact, context, metric, and objective-type artefacts etc. are at the heart of the domain knowledge embedded in a library. These libraries are typically created by hand by humans, who express their knowledge (or others' knowledge) in stylized code such as python.

(230) With the refinement of LLM-based techniques for code generation and generative patterns, it is possible to provide an alternate LLM-based solution for producing libraries. An agent may engage a human expert in conversation and, through a sequence of questions, generate a domain-specific library. This has some parallels with legacy techniques of "knowledge elicitation" but with the advantage of having the ability to express a strong model in code which can then be tested.

(231) This capability empowers non-technical experts to create libraries that can be shared and exploited by other actants without writing any code.

(232) Behavior automation in the present context refers to a process of dynamically associating behaviors with evolving entity models to enable such behaviors to be implemented when appropriate (automatically in response to human input or entirely autonomously).

(233) Object-Oriented Approaches and Dynamic Contexts

(234) Object-oriented approaches build collections of interconnected objects. Object oriented approaches have been used to model complex entities, such as complex and evolving real-world systems.

(235) In object-oriented programming (OOP), an object has a fixed identity and can have a state and one or more methods. Properties, behaviors and relationships to other objects are conventionally defined in the state and method of the objects themselves. OOP is a computer programming model that organizes software design around object identity and behavior, rather than functions and data. In typical OOP contexts, objects defined as instances of classes, where the

classes are linked with functional uses of the classes. Methods are typically predefined in the classes to determine how the objects interact with each other.

(236) In various aspects disclosed herein, an entity is modeled as a collection of artefacts. An explicit modeling approach is adopted, in which artefacts represent particular features of the entity. The model supports behavior binding, whereby the behavior is not defined by the type or identity of the artefact like in OOP systems, but by the pattern of artefact types and pre-conditions. Behavior binding in this sense enables autonomous agents to act based on the model. For example, in one implementation, the entity model serves as a communication model between multiple cooperating autonomous agents that are notified of changes to the entity model, and can propose actions to take in response. This is merely one example, and other forms of behavior automation may be supported.

(237) Rather than binding behaviors in the artefacts themselves, behaviors are identified and instigated based on dynamic context matching. A dynamic context refers to a defined artefact structure—a specific pattern of relationships between defined artefact types along with possible conditions. An available behavior is identified by matching a subset of artefact(s) in the entity model a predefined artefact pattern associated with the behavior. This approach enables specialization and extension of entity models without having to alter artefact definitions. Hence, the dynamic context matching approach provided herein provides greater flexibility and extendibility of behavior based on entity modeling, leading to effective automation of actions within an entity being modelled.

(238) According to an aspect of the present disclosure there is provided a computer-implemented method of automating behaviors performed in relation to a modelled entity, comprising: accessing an entity model comprising a plurality of artefacts, each artefact modelling a feature of the modelled entity; matching a first subset of one or more artefacts in the entity model with a first predefined artefact structure; and in response to matching the first subset of one or more artefacts with the first predefined artefact structure: identifying a first behavior associated with the first predefined artefact structure, initiating the first behavior, and based on the initiated first behavior, adding a new artefact to the entity model, or removing an existing artefact from the entity model, or modifying an existing artefact in the entity model, resulting in an updated entity model.

(239) In some examples, the method comprises matching a second subset of one or more artefacts in the updated entity model with a second predefined artefact structure; and in response to matching the second subset of one or more artefacts with the second predefined artefact structure: identifying a second behavior associated with the second predefined artefact structure, initiating the second behavior, and based on the initiated second behavior, adding a new artefact to the updated entity model, removing an existing artefact from the updated entity model, or modifying an existing artefact of the updated entity model, resulting in a second updated entity model.

(240) In some examples, the first behavior comprises sending a message to an autonomous agent.

(241) In some examples, the method comprises receiving from the autonomous agent a response indicating an action to be performed, the method comprising causing the action to be performed, the updated entity model reflecting the performed action.

(242) In some examples, the method comprises, responsive to matching the first subset of one or more artefacts with the first predefined artefact structure, associating a context artefact with the subset of one or more artefacts of the entity model, wherein the behavior is identified based on the context artefact.

(243) In some examples, the first behavior is defined in a runtime engine external to the context artefact, and is identified in the runtime engine based on a type assigned to the context artefact.

(244) In some examples, the first predefined artefact structure is defined in a context class within a library, wherein the method comprises instantiating, responsive to matching the first subset of one or more artefacts in the entity model with the first predefined artefact structure, the context class, resulting in the context artefact, the context artefact containing a link to at least one artefact of the

first subset of one or more artefacts.

(245) In some examples, each artefact of the entity model is an instance of an artefact class.

(246) In some examples, the method comprises identifying a plurality of available agent engagement behaviors based on the agent artefact in the entity model, wherein the message is sent to the autonomous agent responsive to an input selecting the first behavior from the plurality of available agent engagement behaviors.

(247) In some examples the method comprises, in response to matching the second subset of artefacts with the second predefined artefact structure, associating a second context artefact with the second plurality of one or more artefacts, wherein the second behavior is identified based on the second context artefact.

(248) In some examples, matching the first subset of artefacts in the entity model comprises identifying structural information of each artefact in the first subset of artefacts. Each artefact of the plurality of artefacts in the entity model may comprise structural information defining a feature of an entity being modelled, and defining a structural link between said artefact and one or more other artefact in the entity model.

(249) In some examples, matching the first subset of one or more artefacts with the first predefined artefact structure is performed in response to a change in the entity model.

(250) In some examples, matching the first subset of one or more artefacts with the first predefined artefact structure is performed in response to: a change in a library associated with the entity model, a new library being associated with the entity model, or an existing library being disassociated from the entity model.

(251) In some examples, updating the entity model comprises adding a new artefact, wherein the new artefact contains a link to an existing artefact. The existing artefact may be retained in the updated entity model and may not contain any link to the new artefact.

(252) In accordance with another aspect of the present disclosure, there is provided a system comprising: one or more processor; and computer memory coupled to the one or more processor, and comprising computer executable code which, when executed by the one or more processors causes the one or more processor to: access an entity model comprising a plurality of artefacts, each artefact modelling a feature of the modelled entity; match a first subset of one or more artefacts in the entity model with a first predefined artefact structure; and in response to matching the first subset of one or more artefacts with the first predefined artefact structure: identify a first behavior associated with the first predefined artefact structure; initiate the first behavior, and based on the initiated first behavior, add a new artefact to the entity model, or remove an existing artefact from the entity model, or modify an existing artefact in the entity model, resulting in an updated entity model

(253) In some examples, the computer executable code, when executed by the one or more processors, further causes the one or more processor to: match a second subset of one or more artefacts in the updated entity model with a second predefined artefact structure; and in response to matching the second subset of one or more artefacts with the second predefined artefact structure: identify a second behavior associated with the second predefined artefact structure, initiate the second behavior, and based on the initiated second behavior, add a new artefact to the updated entity model, remove an existing artefact from the updated entity model, or modify an existing artefact of the updated entity model, resulting in a second updated entity model.

(254) In some examples, the computer executable code, when executed by the one or more processors, further causes the one or more processor, responsive to matching the first subset of one or more artefacts with the first predefined artefact structure, to: associate a context artefact with the subset of one or more artefacts of the entity model, wherein the behavior is identified based on the context artefact.

(255) In some examples, the first behavior is defined in a runtime engine external to the context artefact, and is identified in the runtime engine based on a type assigned to the context artefact.

(256) In some examples, the first predefined artefact structure is defined in a context class within a library, and the computer executable code, when executed by the one or more processors, further causes the one or more processor to: responsive to matching the first subset of one or more artefacts in the entity model with the first predefined artefact structure, instantiate the context class, resulting in the context artefact, the context artefact containing a link to at least one artefact of the first subset of one or more artefacts.

(257) In some examples, the computer executable code, when executed by the one or more processors, further causes the one or more processor to: in response to matching the second subset of artefacts with the second predefined artefact structure, associate a second context artefact with the second plurality of one or more artefacts, wherein the second behavior is identified based on the second context artefact.

(258) In accordance with another aspect of the present disclosure there is provided a non-transitory computer-readable medium comprising computer executable instructions which, when executed by one or more processor, cause the one or more processor to implement a method comprising: accessing an entity model comprising a plurality of artefacts, each artefact modelling a feature of the modelled entity; matching a first subset of one or more artefacts in the entity model with a first predefined artefact structure; and in response to matching the first subset of one or more artefacts with the first predefined artefact structure: identifying a first behavior associated with the first predefined artefact structure, initiating the first behavior, and based on the initiated first behavior, adding a new artefact to the entity model, or removing an existing artefact from the entity model, or modifying an existing artefact in the entity model, resulting in an updated entity model.

(259) Further aspects relate to computer-readable instructions for programming a computer system to implement a method according to any embodiment of the aspect described above.

(260) Functional components described herein such as the runtime engine may be implemented as computer-readable instructions stored on a computer-readable medium and executed by one or more processors. The computer-readable medium may include any suitable physical device that stores data in a non-transitory manner. Examples of computer-readable media include, but are not limited to, hard disk drives, solid state drives, flash memory devices, optical discs, magnetic tapes, and the like. The computer-readable medium may also include a transitory or non-transitory signal-bearing medium that stores or transmits data, such as electrical, optical, or electromagnetic signals. The processors that execute the software instructions may be part of any suitable computing device (or system of networked devices), such as a server, a workstation, a laptop, a mobile device, or an embedded system. The processors may include general-purpose or special-purpose processors, such as central processing units (CPUs), graphics processing units (GPUs), application-specific integrated circuits (ASICs), field-programmable gate arrays (FPGAs), or any combination thereof. The processors may be configured to communicate with each other and with other components of the computing device via buses, networks, or other interfaces. Functions and operations described herein may be implemented using any such configuration of hardware.

(261) The foregoing examples are provided for illustration purposes only and should not be construed as limiting the scope of the present disclosure. The scope is defined by the appended claims, rather than by the specific embodiments described above. Those skilled in the art will appreciate that various modifications and variations can be made to the disclosed embodiments without departing from the scope defined in the claims.

Claims

1. A computer-implemented method performed in a central system, the method comprising: forming in electronic storage an initial entity model comprising a plurality of artefacts; determining, by a runtime engine, a plurality of available agent engagements based on the initial entity model, by inspecting the initial entity model against an artefact library held in the electronic

storage and by identifying one or more predefined structures of artefacts in the artefact library matching one or more artefact structures in the initial entity model; responsive to an input selecting one of the available agent engagements, sending a notification to an external autonomous agent hosted on a remote server separate from the central system; receiving from the external autonomous agent an indication of an action to be performed, and responsive thereto: sending a message to an external service indicating the action, and thereby causing the action to be performed, wherein the external service is hosted on a second remote server, and wherein the message comprises a credential causing the external service to authorize the action, the credential held in the central system in a storage location inaccessible to the external autonomous agent, automatically updating the entity model, by the runtime engine, with a new artefact corresponding to the performed action; and determining, by the runtime engine, a further plurality of agent actions based on the updated entity model.

2. The method of claim 1 comprising, in response to determining a further plurality of agent actions: determining, by a runtime engine, a second plurality of available agent engagements based on the updated entity model; responsive to an input selecting one of the available agent engagements of the second plurality of available agent engagements, sending a second notification to a second external autonomous agent; receiving from the second external autonomous agent an indication of a second action to be performed, and responsive thereto: sending a message to a second external service for causing the second action to be performed, and updating the entity model with a second new artefact corresponding to the second performed action.

3. The method of claim 1, wherein each artefact of the plurality of artefacts is generated as an instance of an artefact class defined in the artefact library.

4. The method of claim 3, wherein the plurality of available agent engagements and the further plurality of agent actions are extracted from the artefact library using the initial entity model and the updated entity model respectively.

5. The method of claim 1, wherein the input selecting one of the available agent engagements is received from a user interface based on a selection made by a human user.

6. The method of claim 1, wherein the instruction selecting one of the available agent engagements is received from an external agent engaged to perform a role defined in an artefact of the initial model.

7. The method of claim 6, wherein a first external agent that is notified of a change in the entity model requests engagement of a second agent.

8. The method of claim 6, wherein a first external agent that is notified of a change in the entity model suggests engagement of a second agent to a human user via a user interface.

9. The method of claim 1, wherein determining a further plurality of agent actions based on the updated entity model comprises: inspecting the updated entity model; determining that the updated entity model is in a state that satisfies a criterion defined in the artefact library; and notifying one or more autonomous agent of the state of the entity model.

10. The method of claim 1, wherein the credential comprises an authentication token.

11. A non-transitory computer readable medium storing computer readable instructions which when executed by one or more processor cause the processor to implement operations comprising: forming in electronic storage an initial entity model comprising a plurality of artefacts; determining, by a runtime engine, a plurality of available agent engagements based on the initial entity model, by inspecting the initial entity model against an artefact library held in the electronic storage and by identifying one or more predefined structures of artefacts in the artefact library matching one or more artefact structures in the initial entity model; responsive to an input selecting one of the available agent engagements, sending a notification to an external autonomous agent hosted on a server remote server separate from the central system; receiving from the external autonomous agent an indication of an action to be performed, and responsive thereto: sending a message to an external service indicating the action, and thereby causing the action to be

performed, wherein the external service is hosted on a second remote server, and wherein the message comprises a credential causing the external service to authorize the action, the credential held in the central system in a storage location inaccessible to the external autonomous agent, automatically updating the entity model, by the runtime engine, with a new artefact corresponding to the performed action; and determining, by the runtime engine, a further plurality of agent actions based on the updated entity model.

12. The non-transitory computer readable medium of claim 11, wherein the operations comprise, further in response to determining a further plurality of agent actions: determining, by a runtime engine, a second plurality of available agent engagements based on the updated entity model; responsive to an input selecting one of the available agent engagements of the second plurality of available agent engagements, sending a second notification to a second external autonomous agent; receiving from the second external autonomous agent an indication of a second action to be performed, and responsive thereto: sending a message to a second external service for causing the second action to be performed, and updating the entity model with a second new artefact corresponding to the second performed action.

13. The non-transitory computer readable medium of claim 11, wherein each artefact of the plurality of artefacts is generated as an instance of an artefact class defined in an artefact library.
